Feature Engineering with Generative AI

Mastering AI-driven Feature Engineering

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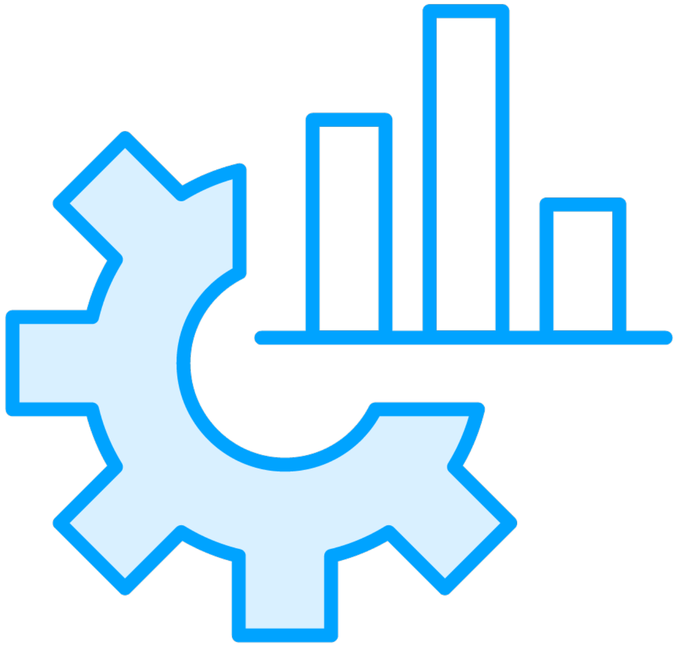
Feature Engineering

Feature engineering is a preprocessing step in supervised machine learning and statistical modeling which transforms raw data into a more effective set of inputs.

https://en.wikipedia.org/wiki/Feature_engineering

Feature Engineering Techniques



Selecting relevant features

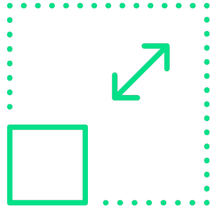
Transforming features

Scaling, encoding, and combining existing features

Creating new features

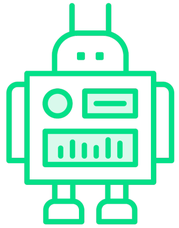
Feature Engineering with Generative AI

Domain knowledge

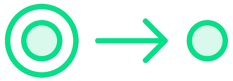


Dimensionality reduction

Data understanding and exploration



Automated feature engineering



Feature transformation and creation

Documentation

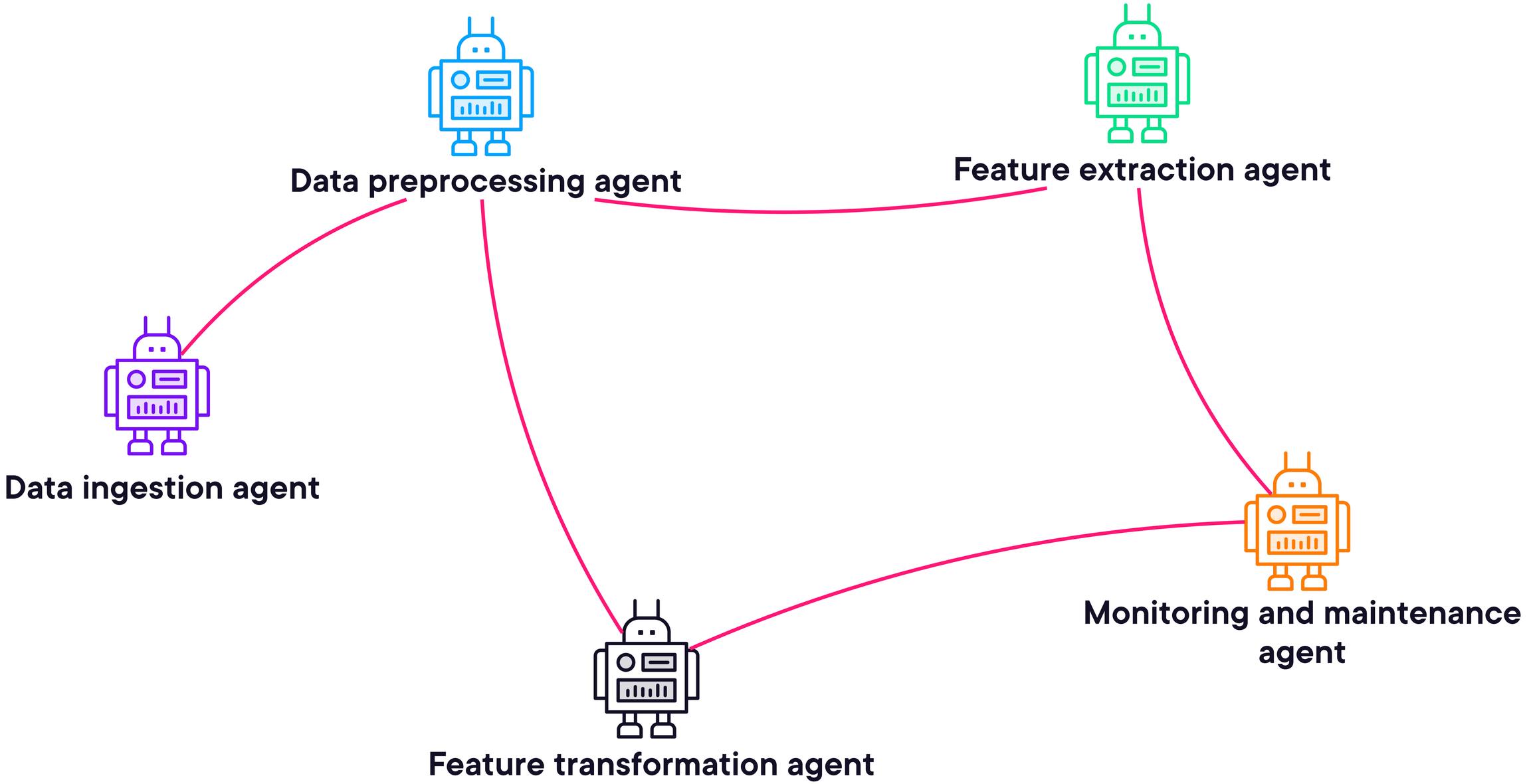
Automating the Feature Engineering Process

Demo

Using APIs with pandas dataframe

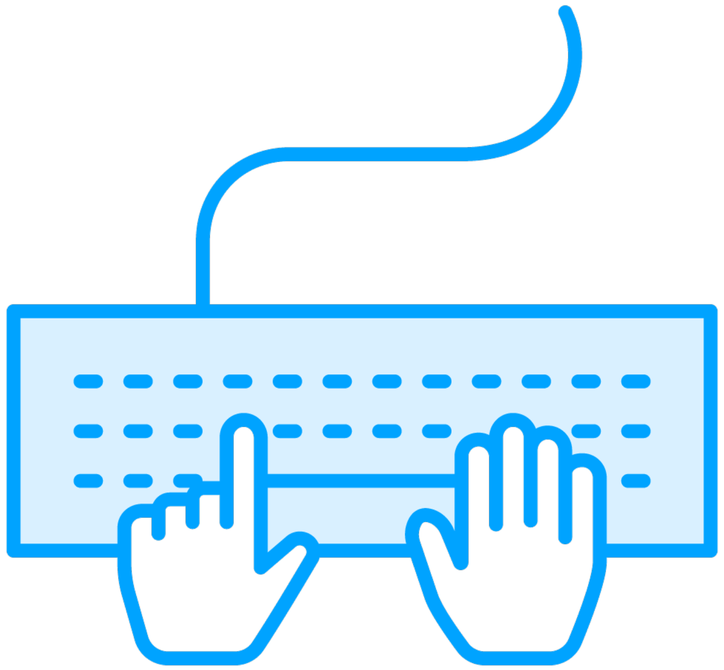
Crafting prompts for feature engineering

Using the knowledge of LLMs


Data preprocessing agent
Feature extraction agent
Data ingestion agent
Monitoring and maintenance agent
Feature transformation agent

Crafting Prompts for Feature Selection

Crafting Prompts for Feature Selection

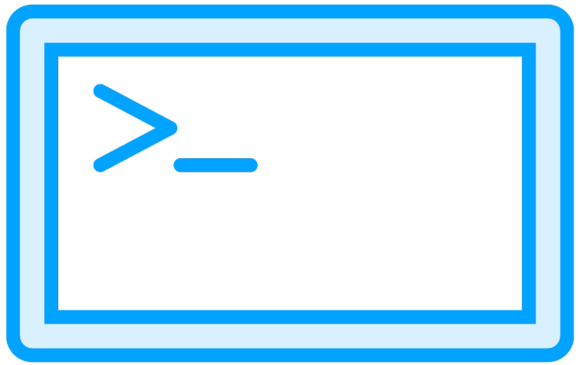


Understand the data

Specify the objective

Constraints and preferences

Prompts for Code Generation



Language and libraries

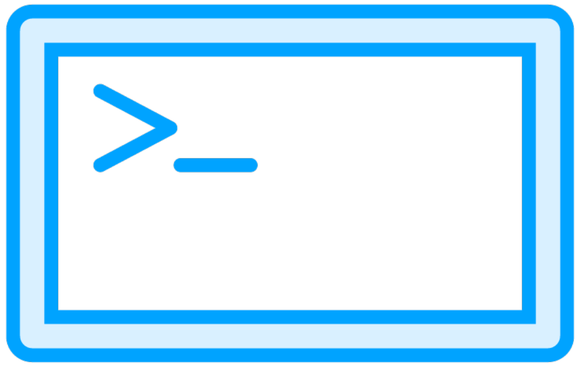
Steps as bullet points

Strategies

Dimensionality Reduction

Assessing and Optimizing Engineered Features

Assessing the Engineered Features



Correlation analysis

Feature importance scores

Baseline scores